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Gas turbine and fuel nozzle assembly for mixing hydrogen fuel and steam to control flame speed

Abstract

A gas turbine engine comprising a compressor section, a combustion section, and a turbine section in a serial flow arrangement, with the combustion section comprising a combustor liner that at least partially defines a combustion chamber; and a gaseous fuel nozzle assembly, comprising: a steam supply to supply steam; a hydrogen fuel supply to supply gaseous hydrogen fuel; and a fuel nozzle body fluidly coupled with the steam supply and the hydrogen fuel supply.

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Background/Summary

TECHNICAL FIELD

(1) The present subject matter relates generally to a gas turbine engine having a fuel nozzle assembly.

BACKGROUND

(2) Turbine engines are driven by a flow of combustion gases passing through the engine to rotate a

multitude of turbine blades, which, in turn, rotate a compressor to provide compressed air to the combustor for combustion. A combustor can be provided within the gas turbine engine and is fluidly coupled with a turbine into which the combusted gases flow.

(3) Historically, hydrocarbon fuels are used in the combustor of a turbine engine. Generally, air and fuel are fed to a combustion chamber, the air and fuel are mixed, and then the fuel is burned in the presence of the air to produce hot gas. The hot gas is then fed to a turbine where it cools and expands to produce power. By-products of the fuel combustion typically include environmentally unwanted byproducts, such as nitrogen oxide and nitrogen dioxide (collectively called NO_x), carbon monoxide (CO), unburned hydrocarbons (UHC) (e.g., methane and volatile organic compounds that contribute to the formation of atmospheric ozone), and other oxides, including oxides of sulfur (e.g., SO₂ and SO₃).

(4) To reduce the environmentally unwanted byproducts, other fuels, such as hydrogen, are being explored. Hydrogen or hydrogen mixed with another element has a higher flame temperature than traditional hydrocarbon fuels. That is, hydrogen or a hydrogen mixed fuel typically has a wider flammable range and a faster burning velocity than traditional hydrocarbon-based fuels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a schematic view of a gas turbine engine having a compression section, a combustion section, and a turbine section in accordance with various aspects described herein.

(3) FIG. 2 is a schematic view of the combustion section of FIG. 1 along line II-II in accordance with various aspects described herein.

(4) FIG. 3 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(5) FIG. 4 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(6) FIG. 5 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(7) FIG. 6 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(8) FIG. 7 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(9) FIG. 8 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(10) FIG. 9 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(11) FIG. 10 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(12) FIG. 11 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(13) FIG. 12 is a schematic cross-sectional view illustrating portions of a fuel nozzle assembly coupled to a combustion chamber in accordance with various aspects described herein.

(14) FIG. 13 is a block diagram depicting an embodiment of a method of operating a gas turbine engine in accordance with various aspects described herein.

DETAILED DESCRIPTION

(15) Aspects of the disclosure described herein are directed to a combustor and a fuel nozzle assembly for a combustor. With some aspects, the disclosed combustors and fuel nozzle assemblies

can be utilized with gaseous fuel, such as hydrogen. Gaseous fuel, including hydrogen, spreads/disperses at a faster rate than atomized liquid fuel, which can involve less mixing time for the gaseous fuel, fuel mixing tube lengths can be shorter, and the flame from the gaseous fuel may be more likely to spread farther and faster, which can increase the risk of flashback and flameholding (e.g., in a nozzle or mixer), and increase the impact of controlling the flame and limiting flame spread by controlling the dispersion of the gaseous fuel.

(16) With fuel nozzle assemblies disclosed herein, mixing gaseous hydrogen with steam can limit flame speeds, which can limit flashback and flameholding. Introduction of steam also reduces peak temperatures and NO_x emissions. Introducing steam around hydrogen fuel can limit radial spread of the flame and reduce wall and liner temperatures. Steam can be mixed with gaseous hydrogen before the fuel nozzle in fluid passages or in a mixing chamber. Steam can be introduced with gaseous hydrogen via one or more orifices inside the fuel nozzle assembly such that steam and gaseous hydrogen mix close to fuel nozzle tip without damaging the fuel nozzle assembly or the swirler. Steam can be injected at different locations along the fuel nozzle body to facilitate mixing with fuel, such as gaseous hydrogen. Steam can be injected around an outermost circuit of the fuel nozzle assembly to control the shape of the flame and to reduce wall and liner temperatures. Steam can be injected at all engine operating conditions or at certain operating conditions to reduce NO_x emissions or to control the shape of the flame for lower wall and liner temperatures, or to limit flashback and flameholding. Steam and gaseous hydrogen can be introduced through one or multiple outlets. Steam can be introduced between fuel outputs to control flame shape and to reduce regions of high temperatures to limit NO_x emissions. Steam can be injected between gaseous hydrogen and air to delay interaction between gaseous hydrogen and air in the low velocity regions close to the wall, which can limit flashback and flameholding.

(17) While described with respect to a turbine engine, it should be appreciated that the combustor as described herein can be for any engine having a combustor. It should be appreciated that application of aspects of the disclosure discussed herein are applicable to engines with propeller sections or fan and booster sections along with turbojets and turbo engines as well.

(18) For purposes of illustration, the present disclosure will be described with respect to a turbine engine. It will be understood, however, that aspects of the disclosure described herein are not so limited. A combustor as described herein can be implemented in various engines, including but not limited to turbojet, turboprop, turboshaft, and turbofan engines. Aspects of the disclosure discussed herein may have general applicability within non-aircraft engines having a combustor, such as other mobile applications and non-mobile industrial, commercial, and residential applications.

(19) With the combustors and fuel nozzle assemblies described herein, gaseous hydrogen fuel can be used without the need of diluents. In some embodiments, no diluent is added to the combustion chamber and the fuel is substantially completely diatomic hydrogen without diluent. As used herein, the term “substantially completely,” as used to describe the amount of a particular element or molecule (e.g., diatomic hydrogen), refers to at least 99% by mass of the described portion of the element or molecule, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the element or molecule. In some examples, the fuel is entirely (e.g., 100%) hydrogen by mass.

(20) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

(21) As used herein, the terms “first”, “second”, “third”, etc. may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

(22) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle

and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine exhaust.

(23) As used herein, the term “upstream” refers to a direction that is opposite the fluid flow direction, and the term “downstream” refers to a direction that is in the same direction as the fluid flow. The term “fore” or “forward” means in front of something and “aft” or “rearward” means behind something. For example, when used in terms of fluid flow, fore/forward can mean upstream and aft/rearward can mean downstream.

(24) The term “fluid” may be a gas or a liquid. The term “fluidly coupled” means that a fluid is capable of making the connection between the areas specified.

(25) The term “nozzle” has been used in various ways in the context of gas turbine engines. In the instant application, “nozzle” refers to a component having a portion for fluid coupling to a fuel supply and having at least one portion for fluidly coupling with a combustor portion, a combustor liner, a combustion chamber, or combinations thereof.

(26) Additionally, as used herein, the terms “radial” or “radially” refer to a direction away from a common center. For example, in the overall context of a turbine engine, radial refers to a direction along a ray extending between a center longitudinal axis of the engine and an outer engine circumference.

(27) All directional references (e.g., radial, axial, proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, upstream, downstream, forward, aft, etc.) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, connected, and joined) are to be construed broadly and can include intermediate structural elements between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only, and the dimensions, positions, order, and relative sizes reflected in the drawings attached hereto can vary.

(28) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. Furthermore, as used herein, the term “set” or a “set” of elements can be any number of elements, including only one.

(29) Uses of “and” and “or” are to be construed broadly. For example and without limitation, uses of “and” do not necessarily require all elements or features listed, and uses of “or” are inclusive unless such a construction would be illogical.

(30) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, “generally”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and systems. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and systems. For example, the approximating language may refer to being within a 1, 2, 4, 5, 10, 15, or 20 percent margin in either individual values, range(s) of values and endpoints defining range(s) of values. Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(31) “Proximate” as used herein is a descriptor for locating parts described herein. Further, the term “proximate” means nearer or closer to the part recited than the following part. For example, a first aperture proximate a wall, the first aperture located upstream from a second aperture means that the first aperture is closer to the wall than the first aperture is to the second aperture.

(32) Additionally, as used herein, a “controller” can include a component configured or adapted to provide instruction, control, operation, or any form of communication for operable components to effect the operation thereof. A controller can include any known processor, microcontroller, or logic device, including, but not limited to: field programmable gate arrays (FPGA), an application specific integrated circuit (ASIC), a full authority digital engine control (FADEC), a proportional controller (P), a proportional integral controller (PI), a proportional derivative controller (PD), a proportional integral derivative controller (PID controller), proportional resonant controller (PR), a hardware-accelerated logic controller (e.g. for encoding, decoding, transcoding, etc.), the like, or a combination thereof. Non-limiting examples of a controller can be configured or adapted to run, operate, or otherwise execute program code to effect operational or functional outcomes, including carrying out various methods, functionality, processing tasks, calculations, comparisons, sensing or measuring of values, or the like, to enable or achieve the technical operations or operations described herein. The operation or functional outcomes can be based on one or more inputs, stored data values, sensed or measured values, true or false indications, or the like. While “program code” is described, non-limiting examples of operable or executable instruction sets can include routines, programs, objects, components, data structures, algorithms, etc., that have the technical effect of performing particular tasks or implement particular abstract data types. In another non-limiting example, a controller can also include a data storage component accessible by the processor, including memory, whether transient, volatile or non-transient, or non-volatile memory.

(33) Additional non-limiting examples of the memory can include Random Access Memory (RAM), Read-Only Memory (ROM), flash memory, or one or more different types of portable electronic memory, such as discs, DVDs, CD-ROMs, flash drives, universal serial bus (USB) drives, the like, or any suitable combination of these types of memory. In one example, the program code can be stored within the memory in a machine-readable format accessible by the processor. Additionally, the memory can store various data, data types, sensed or measured data values, inputs, generated or processed data, or the like, accessible by the processor in providing instruction, control, or operation to effect a functional or operable outcome, as described herein. In another non-limiting example, a controller can be configured for comparing a first value with a second value and operating and controlling operations of additional components based on the satisfying of that comparison. For example, when a sensed, measured, or provided value is compared with another value, including a stored or predetermined value, the satisfaction of that comparison can result in actions, functions, or operations controllable by the controller.

(34) FIG. 1 is a schematic view of a gas turbine engine **10**. As a non-limiting example, the gas turbine engine **10** can be used within an aircraft. The gas turbine engine **10** can include, at least, a compressor section **12**, a combustion section **14**, and a turbine section **16**. A drive shaft **18** rotationally couples the compressor section **12** and turbine section **16**, such that rotation of one affects the rotation of the other, and defines a rotational axis **20** for the gas turbine engine **10**.

(35) The compressor section **12** can include a low-pressure (LP) compressor **22**, and a high-pressure (HP) compressor **24** serially fluidly coupled to one another. The turbine section **16** can include an HP turbine **26**, and an LP turbine **28** serially fluidly coupled to one another. The drive shaft **18** can operatively couple the LP compressor **22**, the HP compressor **24**, the HP turbine **26** and the LP turbine **28** together. Alternatively, the drive shaft **18** can include an LP drive shaft and an HP drive shaft. The LP drive shaft can couple the LP compressor **22** to the LP turbine **28**, and the HP drive shaft can couple the HP compressor **24** to the HP turbine **26**. An LP spool can be defined as the combination of the LP compressor **22**, the LP turbine **28**, and the LP drive shaft such that the rotation of the LP turbine **28** can apply a driving force to the LP drive shaft, which in turn

can rotate the LP compressor **22**. An HP spool can be defined as the combination of the HP compressor **24**, the HP turbine **26**, and the HP drive shaft such that the rotation of the HP turbine **26** can apply a driving force to the HP drive shaft which in turn can rotate the HP compressor **24**.

(36) The compressor section **12** can include a plurality of axially spaced stages. Each stage includes a set of circumferentially-spaced rotating blades and a set of circumferentially-spaced stationary vanes. The compressor blades for a stage of the compressor section **12** can be mounted to a disk, which is mounted to the drive shaft **18**. Each set of blades for a given stage can have its own disk. The vanes of the compressor section **12** can be mounted to a casing which can extend circumferentially about and enshroud one or more sections of the gas turbine engine **10**. It will be appreciated that the representation of the compressor section **12** is merely schematic and that there can be any number of blades, vanes and stages. Further, it is contemplated that there can be any number of other components within the compressor section **12**.

(37) Similar to the compressor section **12**, the turbine section **16** can include a plurality of axially spaced stages, with each stage having a set of circumferentially-spaced, rotating blades and a set of circumferentially-spaced, stationary vanes. The turbine blades for a stage of the turbine section **16** can be mounted to a disk which is mounted to the drive shaft **18**. Each set of blades for a given stage can have its own disk. The vanes of the turbine section **16** can be mounted to the casing in a circumferential manner. It is noted that there can be any number of blades, vanes and turbine stages as the illustrated turbine section **16** is merely a schematic representation. Further, it is contemplated that there can be any number of other components within the turbine section **16**.

(38) The combustion section **14** can be provided serially between the compressor section **12** and the turbine section **16**. The combustion section **14** can be fluidly coupled to at least a portion of the compressor section **12** and the turbine section **16** such that the combustion section **14** at least partially fluidly couples the compressor section **12** to the turbine section **16**. As a non-limiting example, the combustion section **14** can be fluidly coupled to the HP compressor **24** at an upstream end of the combustion section **14** and to the HP turbine **26** at a downstream end of the combustion section **14**. The combustion section **14** can include a combustor **30**.

(39) During operation of the gas turbine engine **10**, ambient or atmospheric air is drawn into the compressor section **12** via a fan (not illustrated) upstream of the compressor section **12**, where the air is compressed defining a pressurized air. The pressurized air can then flow into the combustion section **14** where the pressurized air is mixed with fuel and ignited, thereby generating combustion gases. Some work is extracted from these combustion gases by the HP turbine **26**, which drives the HP compressor **24**. The combustion gases are discharged into the LP turbine **28**, which extracts additional work to drive the LP compressor **22**, and the exhaust gas is ultimately discharged from the gas turbine engine **10** via an exhaust section (not illustrated) downstream of the turbine section **16**. The driving of the LP turbine **28** drives the LP spool to rotate the fan (not illustrated) and the LP compressor **22**. The pressurized airflow and the combustion gases can together define a working airflow that flows through the fan, compressor section **12**, combustion section **14**, and turbine section **16** of the gas turbine engine **10**.

(40) FIG. 2 depicts a cross-sectional view of the combustion section **14** along line II-II of FIG. 1. The combustor **30** can include an annular arrangement of combustor portions **31** disposed around the centerline or rotational axis **20** of the gas turbine engine **10** (e.g., circumferentially spaced from each other in an annular configuration) (FIG. 1). The combustor portions **31** can, in some configurations, include or be configured as combustor cups, fuel cups, or nozzle cups. A fuel nozzle assembly **48** can be connected to each combustor portion **31**. The combustor **30** can have a can, can-annular, or annular arrangement depending on the type of engine in which the combustor **30** is located. In a non-limiting example, the combustor **30** can have a combination arrangement located with a shroud or casing **29** of the gas turbine engine **10** (FIG. 1). The shroud or casing **29** can enshroud or cover at least a portion of the combustion section **14**.

(41) The combustor **30** can be at least partially defined by a combustor liner **40**. In some examples,

the combustor liner **40** can include an outer liner **41** and an inner liner **42** concentric with respect to each other and arranged in an annular fashion about the engine centerline or rotational axis **20**. In some examples, the combustor liner **40** can have an annular structure about the combustor **30**. In some examples, the combustor liner **40** can include multiple segments or portions collectively forming the combustor liner **40**. In some examples, the combustor liner **40** can have an annular structure about the combustor **30**. In some examples, the combustor liner **40** can include multiple segments or portions collectively forming the combustor liner **40**. In some examples, the combustor liner **40** can include the outer liner **41** radially spaced from the inner liner **42**. In some examples, the combustor liner **40** can include a single liner.

(42) The combustor liner **40** can at least partially define a combustion chamber **50** arranged annularly about the rotational axis **20**. For example, a wall **46** (e.g., a dome wall) may be substantially perpendicular to the rotational axis **20** and can cooperate with the outer liner **41**, the inner liner **42**, or both, to at least partially define the combustion chamber **50**. A compressed air passage **32** can be defined at least in part by both the combustor liner **40** and the casing **29**.

(43) The combustor **30** can include or be fluidly coupled to a fuel supply **34** (e.g., a fuel manifold or conduit), a steam supply **36** (e.g., a steam manifold or conduit), or both. The fuel nozzle assembly **48** fluidly couples the fuel supply **34** and the steam supply **36** with one of the combustor portions **31** and the combustion chamber **50**. The fuel nozzle assembly **48** can include a fuel nozzle body **49**. The fuel nozzle assembly **48** is coupled to the wall **46**. A fuel F can include any suitable fuel, including gaseous fuel, such as hydrogen fuel, in non-limiting examples, which can include 100% H₂ (e.g., without a diluent). For example, the fuel nozzle assembly **48** can be a gaseous fuel nozzle assembly, such as a gaseous hydrogen fuel nozzle assembly, and the fuel supply **34** can be a hydrogen fuel supply. In some examples, other fuels can be utilized instead of or with hydrogen. For example, the fuel F can comprise methane, such as in the form of natural gas, highly reactive fuels, or combinations thereof, with or without hydrogen. The combustor portions **31** can be arranged in a circumferentially spaced configuration. The combustor portions **31** can be disposed at a radial distance from the rotational axis **20** that is greater than a radial distance of the inner liner **42** and less than a radial distance of the outer liner **41**. A controller **60** can be connected to and at least partially control operation of the fuel supply **34**, the steam supply **36**, the fuel nozzle assembly **48**, or combinations thereof. The controller **60** can include a processor **62** and a memory **64**. A centerline **33** of the combustion section **14** can be concentric with the rotational axis **20**. The centerline **33** can define a radial direction R, an axial direction A, and a circumferential direction C.

(44) FIG. 3 is a schematic view of an example of one of the combustor portions **31** of the combustor **30**. The combustor portion **31** can be provided, at least in part, by the fuel nozzle assembly **48**. The fuel nozzle assembly **48** can be coupled to the wall **46**, extend into the wall **46**, extend through the wall **46**, or combinations thereof. The fuel nozzle assembly **48** can include a fuel nozzle assembly centerline **35** that is parallel with and radially offset from the centerline **33**. The fuel nozzle assembly centerline **35** can be concentric with a centerline of the combustor portion **31** to which the fuel nozzle assembly **48** is coupled. The fuel nozzle assembly centerline **35** can define a second axial direction A2, a second radial direction R2 perpendicular to the second axial direction A2, and a second circumferential direction C2 about the second axial direction A2. The second axial direction A2 can be parallel to the axial direction A (FIG. 2).

(45) The fuel nozzle body **49** of the fuel nozzle assembly **48** can include a fluid passage **100** extending from a first end **102** (e.g., an outlet) of the fuel nozzle body **49** coupled to the wall **46** to a second end **104** (e.g., an inlet) of the fuel nozzle body **49** disposed opposite the first end **102**. A swirler **110** can be disposed in the fluid passage **100**, such as at the first end **102**. A manifold **120** defining a mixing chamber **122** fluidly coupled with the fluid passage **100** at the second end **104** can be included in the fuel nozzle assembly **48**. The manifold **120** can be separate from or included with the fuel nozzle body **49**. The mixing chamber **122** is fluidly coupled with the fuel supply **34** and the steam supply **36** such that fuel F and steam S mix in the mixing chamber **122** and a

resulting fuel-steam mixture FS is output from the mixing chamber **122** to the fluid passage **100**. The fluid passage **100** fluidly couples the mixing chamber **122** with the combustion chamber **50** such that the fuel-steam mixture FS is provided to the combustion chamber **50**. Mixing fuel F with steam S upstream of the combustion chamber **50**, such as within the mixing chamber **122**, can limit flame speeds, flashback, and flameholding in the fuel nozzle assembly **48**. A ratio of steam S to fuel F in the fuel-steam mixture FS can, for example, be at least 0.01 and less than or equal to 10. The ratio can be controlled by the controller **60** (FIG. 2), such as via one or more valves of the fuel supply **34**, the steam supply **36**, the manifold **120**, or combinations thereof.

(46) Referring to FIGS. 4 and 5, a fuel nozzle assembly **248** that can be utilized in the combustor **30** and coupled to the combustor liner **40**, the wall **46**, or both, is illustrated. The fuel nozzle assembly **248** can include aspects similar to those of the fuel nozzle assembly **48**; therefore, like parts will be described with like numerals further increased by 200, with it being understood that the description of the like parts of the fuel nozzle assembly **48** can apply to the fuel nozzle assembly **248**, except where noted. For example, the fuel nozzle assembly **248** can include a fuel nozzle body **249** and a first fluid passage **300**. The fuel nozzle assembly **248** can provide the fuel-steam mixture FS to the combustion chamber **50**. The fuel nozzle body **249** can include a first end **302** coupled to the wall **46** and a second end **304** opposite the first end **302**. A swirler **310** can be disposed in the first fluid passage **300**, such as at the first end **302**.

(47) Referring to FIG. 4, the first fluid passage **300** can include a first section **303** and a second section **305** fluidly coupled with the first section **303**. The first section **303** can extend from the first end **302** to the second section **305**. The second section **305** can extend from the first section **303** to the second end **304** and be fluidly coupled with the fuel supply **34** to provide fuel F to the first section **303**.

(48) The fuel nozzle assembly **248** can include a second fluid passage **330** that can be at least partially defined by the fuel nozzle body **249**. The second fluid passage **330** can extend from the second end **304** to the first section **303** of the first fluid passage **300**. The second fluid passage **330** can be fluidly coupled with the steam supply **36** to provide steam S from the second end **304** to the first section **303** of the first fluid passage **300**. In some examples, the second fluid passage **330** can include an annular configuration disposed at least partially around the second section **305**. The first section **303** of the first fluid passage **300** can receive fuel F from the second section **305** and steam S from the second fluid passage **330**. Fuel F and steam S can mix together in the first section **303** to form the fuel-steam mixture FS that flows into the combustion chamber **50**. For example, the first section **303** can include a greater cross-sectional area than the second section **305**, the second fluid passage **330**, or both, and the first section **303** can function as a premixing chamber. The swirler **310**, which can be disposed in the first section **303**, can facilitate mixing of fuel F and steam S in the fuel nozzle body **249** upstream of the combustion chamber **50** (e.g., in the first section **303**), which can limit flame speed, limit flashback, and limit flameholding in the fuel nozzle assembly **248**.

(49) Referring to FIG. 5, the second fluid passage **330** of the fuel nozzle assembly **248** can extend from the second end **304** to the first end **302** and can include one or more steam orifices **332** (e.g., internal orifices) fluidly coupling the second fluid passage **330** to the first fluid passage **300** to provide steam S into the first fluid passage **300** at one or more locations along the first fluid passage **300**. The steam orifices **332** can facilitate mixing of fuel F with steam S upstream of the combustion chamber **50**. In some examples, one or more of the steam orifices **332** can be perpendicular to the flow direction of fuel F through the first fluid passage **300**, which can increase penetration of the steam S into the flowing fuel F. Additionally or alternatively, one or more of the steam orifices **332** can be angled downstream, at least to some degree, relative to a flow direction of fuel F through the first fluid passage **300** (e.g., at an angle **333** between directly downstream and perpendicular to the flow direction), which can limit wakes. In nonlimiting examples, the angle **333** can be greater than 0 degrees and less than or equal 45 degrees, or greater than 0 degrees and less

than or equal to 30 degrees, which can increase the concentration of steam S along the surface of the second fluid passage **330** to limit flashback and flameholding. The one or more steam orifices **332** can include a plurality of steam orifices with at least one steam orifice upstream of the swirler **310** (e.g., to facilitate mixing of steam S and fuel F) and at least one steam orifice downstream of the swirler **310** (e.g., for flame shaping). The second fluid passage **330** can include a combustion chamber outlet **334** fluidly coupled with the combustion chamber **50** to emit steam S into the combustion chamber **50**. The combustion chamber outlet **334** can include a plurality of outlets or a single outlet, such as an annular outlet, arranged about an outlet **306** of the first fluid passage **300** disposed at or in the combustion chamber **50**. The combustion chamber outlet **334** of the second fluid passage **330** can emit steam S directly into the combustion chamber **50**, such as for flame shaping. For example, emitting steam S around the fuel-steam mixture FS emitted from the outlet **306** can limit flow of fuel F toward the combustor liner **40** (e.g., radial spread), toward or along the wall **46**, or a combination thereof, which can limit temperatures along the combustor liner **40**, the wall **46**, or both.

(50) Referring to FIGS. **6-9**, a fuel nozzle assembly **448** that can be utilized in the combustor **30** (FIG. **1**) and coupled to the combustor liner **40**, the wall **46**, or both, is illustrated. The fuel nozzle assembly **448** can include aspects similar to those of the fuel nozzle assemblies **48** (FIG. **3**), **248** (FIGS. **4** and **5**); therefore, like parts will be described with like numerals further increased by 200 relative to the fuel nozzle assembly **248** of FIGS. **4** and **5**, with it being understood that the description of the like parts of the fuel nozzle assembly **48**, **248** can apply to the fuel nozzle assembly **448**, except where noted. For example, the fuel nozzle assembly **448** can include a fuel nozzle body **449**, can include a first fluid passage **500**, can include a second fluid passage **530**, and can fluidly couple the fuel supply **34** and the steam supply **36** to the combustion chamber **50** to provide fuel F, steam S, the fuel-steam mixture FS, or combinations thereof to the combustion chamber **50**. The fuel nozzle body **449** can include a first end **502** coupled to the wall **46** and a second end **504** opposite the first end **502**. The first fluid passage **500** includes an outlet **506** at or in the combustion chamber **50**. A first swirler **510** can be disposed in the first fluid passage **500**, such as at the first end **502**.

(51) The fuel nozzle assembly **448** can include an outer wall **540** that can be disposed radially outward of the fuel nozzle body **449**, such as at the first end **502**. The outer wall **540** can be spaced from the first end **502** of the fuel nozzle body **449** such that a first outer fluid passage **542** is formed between the fuel nozzle body **449** and the outer wall **540**. A second swirler **512** can be disposed in the first outer fluid passage **542**. In some examples, the second swirler **512** can mechanically couple the outer wall **540** with the fuel nozzle body **449**. The first outer fluid passage **542** can be fluidly coupled to a source of air **70**, such as the compressor section **12** (FIG. **1**). The fuel nozzle assembly **448** can include a steam conduit **536** fluidly coupled with the second fluid passage **530**. The steam conduit **536** can extend from the second fluid passage **530** radially outward, such as into or through the first outer fluid passage **542**, into or through the second swirler **512**, into or through the outer wall **540**, or a combination thereof.

(52) Referring to FIG. **6**, in some examples, the steam conduit **536** extends radially into the outer wall **540** and extends aft in the second axial direction A2 through the outer wall **540** to emit steam S from the outer wall **540** into the combustion chamber **50**. Steam S emitted from the steam conduit **536** through the outer wall **540** can provide flame shaping and limit outward spread of flames in the combustion chamber **50**, such as to limit temperatures at the combustor liner **40**, the wall **46**, or both. The second fluid passage **530** can include a steam orifice **532** at the first end **502** that fluidly couples the second fluid passage **530** with the first fluid passage **500** upstream of the combustion chamber **50** such that the first fluid passage **500** emits the fuel-steam mixture FS from the outlet **506** into the combustion chamber **50**.

(53) Referring to FIG. **7**, in some examples, the steam conduit **536** extends radially outward from the fuel nozzle body **449** into the first outer fluid passage **542** and includes one or more steam

conduit outlets **538** disposed in the first outer fluid passage **542**. The steam conduit outlets **538** can extend aft, at least to some degree, to emit steam S into the combustion chamber **50**, the first outer fluid passage **542**, or both. Emitting steam S into the first outer fluid passage **542** can facilitate mixing of steam S with air **70**. The steam conduit **536** and the steam conduit outlets **538** can be disposed in the second swirler **512** (e.g., circumferentially between adjacent vanes), forward of the second swirler **512**, aft of the second swirler **512**, or combinations thereof. Emitting steam S into the first outer fluid passage **542** and from the first outer fluid passage **542** into the combustion chamber **50** can reduce NO_x emissions and control the flame position aft of the wall **46**. For example, increasing the amount of steam S mixed with fuel F and air **70** reduces flame speed, which moves the flame farther aft, and the mass of steam S increases the velocity of the mixture, which also pushes the flame farther aft. The output of the first outer fluid passage **542** can be a combination or mixture of steam S and air **70**. The outer wall **540** can include an air passage **544** to provide air **70** (e.g., without steam S or fuel F) to the combustion chamber **50**. The air passage **544** can be radially outward of the first outer fluid passage **542**. The second fluid passage **530** can include the steam orifice **532** fluidly coupling the second fluid passage **530** with the first fluid passage **500** upstream of the combustion chamber **50** such that the first fluid passage **500** emits the fuel-steam mixture FS from the outlet **506** into the combustion chamber **50**.

(54) Referring to FIG. **8**, in some examples, the fuel nozzle assembly **448** can include a combination of the configurations of the steam conduit **536** shown in FIGS. **6** and **7**. For example, the steam conduit **536** can extend radially into and axially through the outer wall **540** to emit steam S through the outer wall **540** into the combustion chamber **50** and can include steam conduit outlets **538** disposed in the first outer fluid passage **542** (e.g., in the second swirler **512**) to emit a mixture of steam S and air **70** from the first outer fluid passage **542** into the combustion chamber **50**.

(55) Referring to FIG. **9**, the second fluid passage **530** can be fluidly coupled to the steam conduit **536** and may not be fluidly coupled with the first fluid passage **500**. For example, the first fluid passage **500** can be fluidly coupled with the fuel supply **34** such that the output from the outlet **506** is fuel F without steam S, and the second fluid passage **530** can provide steam S to the first outer fluid passage **542**, the outer wall **540**, or both. In some examples, the first fluid passage **500** can be fluidly coupled with the fuel supply **34** and the steam supply **36**, such as via a manifold (e.g., manifold **120** in FIG. **3**), such that the output from the outlet **506** is the fuel-steam mixture FS (FIG. **3**).

(56) Referring to FIG. **10**, a fuel nozzle assembly **648** that can be utilized in the combustor **30** (FIG. **1**) and coupled to the combustor liner **40**, the wall **46**, or both, is illustrated. The fuel nozzle assembly **648** can include aspects similar to those of the fuel nozzle assemblies **48** (FIG. **3**), **248** (FIGS. **4** and **5**), **448** (FIGS. **6-9**); therefore, like parts will be described with like numerals further increased by 200 relative to the fuel nozzle assembly **448** of FIGS. **6-9**, with it being understood that the description of the like parts of the fuel nozzle assembly **48**, **248**, **448** can apply to the fuel nozzle assembly **648**, except where noted. For example, the fuel nozzle assembly **648** can include a fuel nozzle body **649**, can include a first fluid passage **700**, can include a second fluid passage **730**, can include an outer wall **740**, and can fluidly couple the fuel supply **34** and the steam supply **36** to the combustion chamber **50** to provide fuel F and steam S to the combustion chamber **50**. The fuel nozzle body **649** can include a first end **702** coupled to the wall **46** and a second end **704** opposite the first end **702**. The first fluid passage **700** can include an outlet **706** at or in the combustion chamber **50**. A first swirler **710** can be disposed in the first fluid passage **700**, such as at the first end **702**.

(57) The second fluid passage **730** can include a combustion chamber outlet **734** fluidly coupling the second fluid passage **730** with the combustion chamber **50**. In some examples, the second fluid passage **730** is not fluidly coupled with the first fluid passage **700** such that the first fluid passage **700** emits fuel F without steam S through the outlet **706** into the combustion chamber **50**. The outer wall **740** can radially spaced from first end **702** of the fuel nozzle body **649** such that a first outer

fluid passage **742** is defined between the fuel nozzle body **649** and the outer wall **740**. A second swirler **712** can be disposed in the first outer fluid passage **742**. The first outer fluid passage **742** can be fluidly coupled with a source of air **70**, such as the compressor section **12** (FIG. 1). In some examples, the first outer fluid passage **742** can include a portion that extends (e.g., radially) through the outer wall **740**, such as for fluid coupling with the source of air **70**.

(58) The first fluid passage **700** can include a segment **743** that extends into and through the outer wall **740** (e.g., in the second axial direction **A2**) to emit fuel **F** through the outer wall **740** into the combustion chamber **50**. For example, the segment **743** can provide the first fluid passage **700** with a second outlet **707** fluidly coupled with the combustion chamber **50**. The segment **743** can be spaced radially outward from the first end **702** of the fuel nozzle body **649**. The segment **743** can include a single segment (e.g., an annular segment) or a plurality of segments, such as circumferentially spaced segments disposed in a circumferential row.

(59) The second fluid passage **730** can include a segment **744** that extends into and through the outer wall **740** (e.g., in the second axial direction **A2**) to emit steam **S** through the outer wall **740** into the combustion chamber **50**. For example, the segment **744** can provide the second fluid passage **730** with a second combustion chamber outlet **735** fluidly coupled with the combustion chamber **50**. The segment **744** can be spaced radially outward from a first end **702** of the fuel nozzle body **649**. The segment **744** can include a single segment or a plurality of segments, such as circumferentially spaced segments in one or more circumferential rows. The segment **744** can be disposed about the segment **743**. For example, the segment **743** can include a single annular segment or a single circumferential row of segments (e.g., a middle circle), and the segment **744** can include a first annular segment or circumferential row **745** disposed radially outward of the segment **743** (e.g., an outer circle) and can include a second annular segment or circumferential row **746** disposed radially inward of the segment **744** (e.g., an inner circle). Disposing the segment **744** at both radial sides of the segment **743** can increase axial penetration of fuel **F** into the combustion chamber **50**, which can limit fuel **F** in the combustion chamber **50** at or near the wall **46**, the fuel nozzle assembly **648**, or both. The outer wall **740** can be tapered such that the first annular segment or circumferential row **745** extends farther axially (e.g., into the combustion chamber **50**) than the segment **744**, which can extend farther axially than the second annular segment or circumferential row **746**.

(60) The combustion chamber outlet **734** of the second fluid passage **730** can be disposed radially between the outlet **706** and the first outer fluid passage **742** such that steam **S** is provided between fuel **F** from the outlet **706** and air **70** from the first outer fluid passage **742**, which can delay interaction between fuel **F** and air **70** in low velocity regions of the combustion chamber **50** near the wall **46**, which can limit flashback, flameholding, and temperatures. The second combustion chamber outlet **735** (e.g., at the second annular segment or circumferential row **746**) can be disposed, at least partially, radially between the first outer fluid passage **742** and the second outlet **707** such that steam **S** is provided between fuel **F** from the second outlet **707** and air **70** from the first outer fluid passage **742**, which can delay interaction between fuel **F** and air **70** in low velocity regions of the combustion chamber **50** near the wall **46**, which can limit flashback, flameholding, and temperatures. The fuel nozzle assembly **648** can facilitate mixing steam **S** and fuel **F** aft of the fuel nozzle assembly **648** before interacting with air **70**, which can reduce flame speeds. In some nonlimiting examples, fuel **F** and steam **S** are not mixed within the fuel nozzle assembly **648**.

(61) Referring to FIGS. 11 and 12, a fuel nozzle assembly **848** that can be utilized in the combustor **30** (FIG. 1) and coupled to the combustor liner **40**, the wall **46**, or both, is illustrated. The fuel nozzle assembly **848** can include aspects similar to those of the fuel nozzle assemblies **48** (FIG. 3), **248** (FIGS. 4 and 5), **448** (FIGS. 6-9), **648** (FIG. 10); therefore, like parts will be described with like numerals further increased by 200 relative to the fuel nozzle assembly **648** of FIG. 10, with it being understood that the description of the like parts of the fuel nozzle assembly **48**, **248**, **448**, **648** can apply to the fuel nozzle assembly **848**, except where noted. For example, the fuel nozzle

assembly **848** can include a fuel nozzle body **849**, can include a first fluid passage **900**, can include an outer wall **940**, and can fluidly couple the fuel supply **34** and the steam supply **36** to the combustion chamber **50**. The fuel nozzle body **849** can include a first end **902** coupled to the wall **46** and a second end **904** opposite the first end **902**. The first fluid passage **900** includes a first outlet **906**. A first swirler **910** can be disposed in the first fluid passage **900**, such as at the first end **902**. The first fluid passage **900** can be fluidly coupled to the fuel supply **34** and the steam supply **36**, such as via a manifold **920** that includes a mixing chamber **922**, to emit the fuel-steam mixture FS into the combustion chamber **50**.

(62) The fuel nozzle assembly **848** can include a first intermediate wall **950** and a second intermediate wall **952**. The first intermediate wall **950** can be disposed, at least partially, radially between the fuel nozzle body **849** (e.g., the first end **902**) and the outer wall **940** such that a first outer fluid passage **942** is defined radially between the first intermediate wall **950** and the fuel nozzle body **849**. A second swirler **912** can be disposed in the first outer fluid passage **942**. The second intermediate wall **952** can be disposed, at least partially, radially between the first intermediate wall **950** and the outer wall **940** such that a second outer fluid passage **954** is defined, at least partially, radially between the first intermediate wall **950** and the second intermediate wall **952**, and a third outer fluid passage **956** is defined, at least partially, radially between the second intermediate wall **952** and the outer wall **940**. A third swirler **914** can be disposed in the second outer fluid passage **954**. A fourth swirler **916** can be disposed in the third outer fluid passage **956**. The first intermediate wall **950** can act as a splitter to split air **70** between the first outer fluid passage **942** and the second outer fluid passage **954**. The second intermediate wall **952** can act as a splitter to split air **70** between the second outer fluid passage **954** and the third outer fluid passage **956**. The wall **46** can include a wall air passage **958** that extends through the wall **46** to emit air **70** into the combustion chamber **50**, such as to provide flame shaping and limit the flow of fuel F toward the combustor liner **40**, the wall **46**, or both, which can limit flashback, flameholding, and temperatures. The wall air passage **958** can be angled radially inward toward a fuel nozzle assembly centerline **835** of the fuel nozzle assembly **848**, at least to some extent. The wall air passage **958** can comprise a plurality of wall air passages or a single annular air passage, for example. The fuel nozzle body **849** can include an air passage **959** that extends through at least a portion of the first fluid passage **900** to emit air **70** into the combustion chamber **50**. For example, the air passage **959** can be aligned with a center of the first fluid passage **900** in the first end **902** of the fuel nozzle body **849** and emit air **70** at the center of the first outlet **906**, such as along the fuel nozzle assembly centerline **835**. The air passage **959** can emit air **70** to push recirculation of fuel F away from the fuel nozzle assembly **848** and the wall **46**, which can limit flame propagation toward the fuel nozzle assembly **848** and the wall **46** and limit wear.

(63) Referring to FIG. **11**, the first fluid passage **900** can include a first segment **960** that extends from the fuel nozzle body **849** into and through (e.g., axially) the first intermediate wall **950** to emit the fuel-steam mixture FS into the combustion chamber **50**. For example, the first segment **960** can provide the first fluid passage **900** with a second outlet **907** to emit the fuel-steam mixture FS into the combustion chamber **50**. The first fluid passage **900** can include a second segment **962** that extends from the fuel nozzle body **849** into and through (e.g., axially) the second intermediate wall **952** to emit the fuel-steam mixture FS in the combustion chamber **50**. For example, the second segment **962** can provide the first fluid passage **900** with a third outlet **908** to emit the fuel-steam mixture FS into the combustion chamber **50**. Emitting the fuel-steam mixture FS from radially spaced outlets (e.g., multiple circuits), such as the first outlet **906**, the second outlet **907**, and the third outlet **908**, can more evenly distribute fuel F and steam S, which can provide lower flame speeds to limit flashback and flameholding. While the first segment **960** and the second segment **962** are shown as fluidly coupled to a common portion of the first fluid passage **900**, in some examples, the first fluid passage **900** can include separate portions for providing the fuel-steam mixture FS to the first segment **960** and the second segment **962** (e.g., from the manifold **920**),

which can allow for the first outlet **906**, the second outlet **907**, and the third outlet **908** to be independently controlled (e.g., amount of the fuel-steam mixture FS, steam to fuel ratio of the fuel-steam mixture FS, etc.), such as via the manifold **920**. Increasing the steam S in the fuel-steam mixture FS can reduce temperatures in the combustion chamber **50**, which can reduce NO.sub.x emissions.

(64) Referring to FIG. **12**, the fuel nozzle assembly **848** (e.g., the fuel nozzle body **849**) can include a second fluid passage **930**. The second fluid passage **930** can be fluidly coupled with the steam supply **36** and can include a second fluid passage segment **964** that extends into and through the first intermediate wall **950** to a second fluid passage outlet **965** to emit steam S into the combustion chamber **50**. The fuel nozzle assembly **848** can include the second fluid passage segment **964** instead of the first segment **960** (FIG. **11**) of the first fluid passage **900**. Alternatively, the fuel nozzle assembly **848** can include the second fluid passage segment **964** in addition to the first segment **960** of FIG. **11** (e.g., the second fluid passage segment **964** and the first segment **960** of FIG. **11** can be circumferentially offset). In some examples, the second fluid passage segment **964** can extend through the second intermediate wall **952**, such as instead of or in addition to the second segment **962**. The second fluid passage segment **964** can emit steam S (e.g., without fuel F) radially between the fuel-steam mixture FS emitted from the first outlet **906** and from the third outlet **908**, which can reduce temperatures and NO.sub.x emissions.

(65) Referring to FIG. **13**, a method **1000** of operating a gas turbine engine is illustrated. The method **1000** can be utilized with the gas turbine engine **10** and components thereof, such as those generally illustrated in FIGS. **1-12**. The method **1000** can comprise providing fuel F, such as gaseous hydrogen fuel, to a fuel nozzle assembly (block **1002**) (e.g., a gaseous fuel nozzle assembly), which can include one or more of the fuel nozzle assemblies **48** (FIG. **3**), **248** (FIGS. **4** and **5**), **448** (FIGS. **6-9**), **648** (FIG. **10**), **848** (FIGS. **11** and **12**), fluidly coupled with the combustion chamber **50**. The method **1000** can include providing steam S to the fuel nozzle assembly **48**, **248**, **448**, **648**, **848** (block **1004**). The method **1000** can include controlling a flame speed of a flame in the combustion chamber **50** by providing steam S from the fuel nozzle assembly **48**, **248**, **448**, **648**, **848** to the combustion chamber **50** (block **1006**). Controlling the flame speed can, for example, include limiting the flame speed to a maximum speed of 6 meters per second, which can limit flashback. Controlling the flame speed can, for example, include limiting the flame speed to a maximum speed of 3 meters per second, which can further limit flashback. In some nonlimiting examples, the fuel F can be 100% hydrogen gas, the steam S can be 100% water vapor, or both. Providing steam S from the fuel nozzle assembly **48**, **248**, **448**, **648**, **848** to the combustion chamber **50** (FIGS. **2-12**) (block **1004**) can include emitting steam S into a first outer fluid passage **542** (FIGS. **6-9**) of the fuel nozzle assembly **448** to mix steam S with air **70** (FIGS. **6-9**) prior to mixing with fuel F, such as via one or more steam conduit outlets **538** (FIGS. **6-9**) that can, for example, be disposed in a swirler (e.g., the second swirler **512** (FIGS. **6-9**)). Optionally, the method **1000** can include mixing steam S and fuel F in a mixing chamber **122** (FIG. **3**), **922** (FIGS. **11** and **12**) fluidly coupled to an inlet of the fuel nozzle assembly **48**, **248**, **448**, **648**, **848** (block **1008**). The method **1000** can include shaping the flame with steam S from the fuel nozzle assembly **48**, **248**, **448**, **648**, **848** (block **1010**), which can include limiting the radial spread of fuel F by providing steam S from the fuel nozzle assembly **48**, **248**, **448**, **648**, **848** around fuel F from the fuel nozzle assembly **48**, **248**, **448**, **648**, **848**.

(66) The amount of steam S provided to the combustion chamber **50**, such as via the steam supply **36** (FIGS. **2-12**), the manifolds **120** (FIG. **3**), **920** (FIGS. **11** and **12**), or combinations thereof, can be controlled according to engine conditions, such as via the controller **60** (FIG. **2**).

(67) While second fluid passages, such as second fluid passages **330** (FIGS. **4** and **5**), **530** (FIGS. **6-9**), **730** (FIG. **10**), **930** (FIG. **12**), are described as fluidly coupled with the steam supply **36**, the second fluid passages **330**, **530**, **730**, **930** can, at least in some examples, also be fluidly coupled to the fuel supply **34**, such as to provide a mixture of steam S and fuel F.

(68) In some examples the same combustor **30** (FIGS. 2-5) can include a combination of some or all of the different configurations of the fuel nozzle assemblies **48, 248, 448, 648, 848**. The configurations of the fuel nozzle assemblies **48, 248, 448, 648, 848** are not mutually exclusive, unless expressly described as such, and can be combined in various manners in the same or different combustors.

(69) To the extent not already described, the different features and structures of the various embodiments can be used in combination, or in substitution with each other as desired. That one feature is not illustrated in all of the embodiments is not meant to be construed that it cannot be so illustrated but is done for brevity of description. Thus, the various features of the different embodiments can be mixed and matched as desired to form new embodiments, whether or not the new embodiments are expressly described. All combinations or permutations of features described herein are covered by this disclosure.

(70) This written description uses examples to describe aspects of the disclosure described herein, including the best mode, and also to enable any person skilled in the art to practice aspects of the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of aspects of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(71) Further aspects are provided by the subject matter of the following clauses:

(72) A gas turbine engine, comprising: a compressor section, combustion section, and turbine section in serial flow arrangement, with the combustion section comprising: a combustor liner that at least partially defines a combustion chamber; and a gaseous fuel nozzle assembly, comprising: a steam supply to supply steam; a hydrogen fuel supply to supply gaseous hydrogen fuel; and a fuel nozzle body fluidly coupled with the steam supply and the hydrogen fuel supply to provide the steam and the gaseous hydrogen fuel to the combustion chamber to limit a flame speed in the combustion chamber.

(73) The gas turbine engine of any preceding clause, wherein the gaseous fuel nozzle assembly includes a manifold with a mixing chamber fluidly coupled with the fuel nozzle body, the mixing chamber configured to mix the steam with the gaseous hydrogen fuel.

(74) The gas turbine engine of any preceding clause, wherein the fuel nozzle body includes a first fluid passage fluidly coupled with the steam supply and the combustion chamber, and a second fluid passage fluidly coupled with the steam supply and at least one of the combustion chamber or the first fluid passage.

(75) The gas turbine engine of any preceding clause, wherein the second fluid passage includes a plurality of orifices fluidly coupled with the first fluid passage to permit the steam to flow from the second fluid passage into the first fluid passage to mix with the gaseous hydrogen fuel in the first fluid passage.

(76) The gas turbine engine of any preceding clause, wherein the gaseous fuel nozzle assembly includes a swirler in the first fluid passage; and wherein the plurality of orifices is upstream of the swirler.

(77) The gas turbine engine of any preceding clause, wherein the gaseous fuel nozzle assembly includes a swirler in the first fluid passage; and wherein the plurality of orifices includes at least one orifice upstream of the swirler to facilitate mixing of the steam with the gaseous hydrogen fuel, and at least one orifice downstream of the swirler for flame shaping.

(78) The gas turbine engine of any preceding clause, wherein the gaseous fuel nozzle assembly includes a steam conduit fluidly coupled with the second fluid passage and extending radially outward from the fuel nozzle body.

(79) The gas turbine engine of claim any preceding clause, wherein the gaseous fuel nozzle

assembly includes an outer wall radially spaced from a first end of the fuel nozzle body such that a first outer fluid passage is at least partially defined between the outer wall and the fuel nozzle body.

(80) The gas turbine engine of any preceding clause, wherein the steam conduit extends radially outward into the outer wall and axially aft through the outer wall to emit steam into the combustion chamber.

(81) The gas turbine engine of any preceding clause, wherein the steam conduit includes a plurality of steam conduit outlets disposed in the first outer fluid passage.

(82) The gas turbine engine of any preceding clause, wherein the gaseous fuel nozzle assembly includes a swirler disposed in the first outer fluid passage.

(83) The gas turbine engine of any preceding clause, wherein the outer wall includes an air passage to emit air into the combustion chamber.

(84) The gas turbine engine of any preceding clause, wherein the fuel nozzle body includes a first fluid passage fluidly coupled with the steam supply and the combustion chamber, and a second fluid passage fluidly coupled with the steam supply and at least one of the combustion chamber or the first fluid passage; wherein the gaseous fuel nozzle assembly includes an outer wall radially spaced from a first end of the fuel nozzle body such that a first outer fluid passage is at least partially defined between the outer wall and the fuel nozzle body; wherein the first fluid passage includes a first segment that extends through the outer wall to emit the gaseous hydrogen fuel into the combustion chamber; and wherein the second fluid passage includes a second segment that extends through the outer wall and at least partially around the first segment to emit the steam into the combustion chamber.

(85) The gas turbine engine of any preceding clause, wherein the gaseous fuel nozzle assembly includes: an outer wall radially spaced from a first end of the fuel nozzle body, a first intermediate wall disposed at least partially between the outer wall and the nozzle body, a second intermediate wall disposed at least partially between the first intermediate wall and the outer wall, and a first fluid passage fluidly coupled with at least one of the hydrogen fuel supply and the steam supply; and wherein the first fluid passage includes a first segment extending through the second intermediate wall to the combustion chamber.

(86) The gas turbine engine of any preceding clause, wherein the first fluid passage is fluidly coupled with the hydrogen fuel supply and the steam supply via a manifold; wherein the gaseous fuel nozzle assembly includes a second fluid passage fluidly coupled with the steam supply; and wherein the second fluid passage includes a second segment extending through the first intermediate wall to emit the steam into the combustion chamber.

(87) The gas turbine engine of any preceding clause, wherein the gaseous hydrogen fuel is 100% hydrogen gas.

(88) A method of operating a gas turbine engine having a compressor section, a combustion section including a combustion chamber, and a turbine section in a serial flow arrangement, the method comprising: providing gaseous hydrogen fuel to a gaseous fuel nozzle assembly fluidly coupled with the combustion chamber; providing steam to the gaseous fuel nozzle assembly; and controlling a flame speed of a flame in the combustion chamber by providing the steam from the gaseous fuel nozzle assembly to the combustion chamber.

(89) The method of any preceding clause, wherein the controlling the flame speed includes limiting the flame speed to a maximum speed of 6 meters per second.

(90) The method of any preceding clause, wherein the controlling the flame speed includes limiting the flame speed to a maximum speed of 3 meters per second.

(91) The method of any preceding clause, wherein providing the steam from the gaseous fuel nozzle assembly to the combustion chamber includes emitting the steam into a first outer fluid passage of the fuel nozzle assembly to mix the steam with air prior to mixing with the gaseous hydrogen fuel.

(92) The method of any preceding clause, wherein the gaseous hydrogen fuel is 100% hydrogen

gas.

(93) The method of claim any preceding clause, further comprising mixing the steam and the gaseous hydrogen fuel in a mixing chamber fluidly coupled to an inlet of the gaseous fuel nozzle assembly.

(94) The method of any preceding clause, further comprising shaping the flame with the steam from the gaseous fuel nozzle assembly.

(95) The method of any preceding clause, further comprising limiting radial spread of the flame by providing the steam from the gaseous fuel nozzle assembly around the gaseous hydrogen fuel from the gaseous fuel nozzle assembly.

(96) A method of operating a gas turbine engine having a compressor section, a combustion section including a combustion chamber, and a turbine section in a serial flow arrangement, the method comprising: providing fuel to a fuel nozzle assembly fluidly coupled with the combustion chamber; providing steam to the gaseous fuel nozzle assembly; and controlling a flame speed of a flame in the combustion chamber by providing the steam from the gaseous fuel nozzle assembly to the combustion chamber.

(97) A gaseous fuel nozzle assembly, comprising: a steam supply to supply steam; a hydrogen fuel supply to supply gaseous hydrogen fuel; and a fuel nozzle body fluidly coupled with the steam supply and the hydrogen fuel supply to provide the steam and the gaseous hydrogen fuel to a combustion chamber.

(98) The gaseous fuel nozzle assembly of any preceding clause, further comprising a manifold with a mixing chamber fluidly coupled with the fuel nozzle body, the mixing chamber configured to mix the steam with the gaseous hydrogen fuel.

(99) The gaseous fuel nozzle assembly of any preceding clause, wherein the fuel nozzle body includes a first fluid passage fluidly coupled with the steam supply and the combustion chamber, and a second fluid passage fluidly coupled with the steam supply and at least one of the combustion chamber or the first fluid passage.

(100) The gaseous fuel nozzle assembly of any preceding clause, wherein the second fluid passage includes a plurality of orifices fluidly coupled with the first fluid passage to permit the steam to flow from the second fluid passage into the first fluid passage to mix with the gaseous hydrogen fuel in the first fluid passage.

(101) The gaseous fuel nozzle assembly of any preceding clause, further comprising a swirler in the first fluid passage; and wherein the plurality of orifices is upstream of the swirler.

(102) The gaseous fuel nozzle assembly of any preceding clause, further comprising a swirler in the first fluid passage; wherein the plurality of orifices includes at least one orifice upstream of the swirler to facilitate mixing of the steam with the gaseous hydrogen fuel, and at least one orifice downstream of the swirler for flame shaping.

(103) The gaseous fuel nozzle assembly of any preceding clause, further comprising a steam conduit fluidly coupled with the second fluid passage and extending radially outward from the fuel nozzle body.

(104) The gaseous fuel nozzle assembly of any preceding clause, further comprising an outer wall radially spaced from a first end of the fuel nozzle body such that a first outer fluid passage is at least partially defined between the outer wall and the fuel nozzle body.

(105) The gaseous fuel nozzle assembly of any preceding clause, wherein the steam conduit extends radially outward into the outer wall and axially aft through the outer wall to emit steam into the combustion chamber.

(106) The gaseous fuel nozzle assembly of any preceding clause, wherein the fluid conduit includes a plurality of steam conduit outlets disposed in the first outer fluid passage.

(107) The gaseous fuel nozzle assembly of any preceding clause, wherein the further comprising a swirler disposed in the first outer fluid passage.

(108) The gaseous fuel nozzle assembly of any preceding clause, wherein the outer wall includes

an air passage to emit air into the combustion chamber.

(109) The gaseous fuel nozzle assembly of any preceding clause, wherein the fuel nozzle body includes a first fluid passage fluidly coupled with the steam supply and the combustion chamber, and a second fluid passage fluidly coupled with the steam supply and at least one of the combustion chamber or the first fluid passage; wherein the gaseous fuel nozzle assembly includes an outer wall radially spaced from a first end of the fuel nozzle body such that a first outer fluid passage is at least partially defined between the outer wall and the fuel nozzle body; wherein the first fluid passage includes a first segment that extends through the outer wall to emit the gaseous hydrogen fuel into the combustion chamber; and wherein the second fluid passage includes a second segment that extends through the outer wall and at least partially around the first segment to emit the steam into the combustion chamber.

(110) The gaseous fuel nozzle assembly of any preceding clause, further comprising an outer wall radially spaced from a first end of the fuel nozzle body, a first intermediate wall disposed at least partially between the outer wall and the nozzle body, a second intermediate wall disposed at least partially between the first intermediate wall and the outer wall, and a first fluid passage fluidly coupled with at least one of the fuel supply and the steam supply; wherein the first fluid passage includes a segment extending through the second intermediate wall to the combustion chamber.

(111) The gaseous fuel nozzle assembly of any preceding clause, wherein the first fluid passage is fluidly coupled with the hydrogen fuel supply and the steam supply via a manifold; wherein gaseous fuel nozzle assembly includes a second fluid passage fluidly coupled with the steam supply; and wherein the second fluid passage includes a segment extending through the first intermediate wall to emit the steam into the combustion chamber.

Claims

1. A gas turbine engine, comprising: a compressor section, combustion section, and turbine section in serial flow arrangement, with the combustion section comprising: a combustor liner that at least partially defines a combustion chamber; and a gaseous fuel nozzle assembly, comprising: a steam supply to supply steam; a hydrogen fuel supply to supply gaseous hydrogen fuel; and a fuel nozzle body fluidly coupled with the steam supply and the hydrogen fuel supply to provide the steam and the gaseous hydrogen fuel to the combustion chamber to limit a flame speed in the combustion chamber; wherein the fuel nozzle body includes a first fluid passage and a second fluid passage; wherein the first fluid passage is fluidly coupled with the hydrogen fuel supply; wherein the second fluid passage is fluidly coupled with the steam supply and includes an annular configuration disposed at least partially around the first fluid passage; and wherein the second fluid passage includes a plurality of orifices fluidly coupled with the first fluid passage to enable the steam to flow from the second fluid passage into the first fluid passage to mix with the gaseous hydrogen fuel in the first fluid passage.

2. The gas turbine engine of claim 1, wherein the first fluid passage is fluidly coupled with the combustion chamber, and the second fluid passage is fluidly coupled with at least one of the combustion chamber or the first fluid passage.

3. The gas turbine engine of claim 2, wherein the gaseous fuel nozzle assembly includes a steam conduit fluidly coupled with the second fluid passage and extending radially outward from the fuel nozzle body.

4. The gas turbine engine of claim 3, wherein the gaseous fuel nozzle assembly includes an outer wall radially spaced from a first end of the fuel nozzle body such that a first outer fluid passage is at least partially defined between the outer wall and the fuel nozzle body.

5. The gas turbine engine of claim 4, wherein the steam conduit includes a plurality of steam conduit outlets disposed in the first outer fluid passage.

6. The gas turbine engine of claim 5, wherein the gaseous fuel nozzle assembly includes a swirler

disposed in the first outer fluid passage.

7. The gas turbine engine of claim 6, wherein the outer wall includes an air passage to emit air into the combustion chamber.

8. The gas turbine engine of claim 1, wherein the gaseous fuel nozzle assembly includes a swirler in the first fluid passage; and wherein the plurality of orifices is upstream of the swirler.

9. The gas turbine engine of claim 1, wherein the gaseous fuel nozzle assembly includes a swirler in the first fluid passage; and wherein the plurality of orifices includes at least one orifice upstream of the swirler to facilitate mixing of the steam with the gaseous hydrogen fuel, and at least one orifice downstream of the swirler for flame shaping.

10. The gas turbine engine of claim 1, wherein the gaseous fuel nozzle assembly includes: an outer wall radially spaced from a first end of the fuel nozzle body, a first intermediate wall disposed at least partially between the outer wall and the fuel nozzle body, a second intermediate wall disposed at least partially between the first intermediate wall and the outer wall, and a first fluid passage fluidly coupled with at least one of the hydrogen fuel supply and the steam supply; and wherein the first fluid passage includes a first segment extending through the second intermediate wall to the combustion chamber.

11. The gas turbine engine of claim 1, wherein the gaseous hydrogen fuel is 100% hydrogen gas.

12. A gas turbine engine, comprising: a compressor section, combustion section, and turbine section in serial flow arrangement, with the combustion section comprising: a combustor liner that at least partially defines a combustion chamber; and a gaseous fuel nozzle assembly, comprising: a steam supply to supply steam; a hydrogen fuel supply to supply gaseous hydrogen fuel; and a fuel nozzle body fluidly coupled with the steam supply and the hydrogen fuel supply to provide the steam and the gaseous hydrogen fuel to the combustion chamber to limit a flame speed in the combustion chamber; wherein the fuel nozzle body includes a first fluid passage and a second fluid passage; wherein the first fluid passage is fluidly coupled with the hydrogen fuel supply; wherein the second fluid passage is fluidly coupled with the steam supply and includes an annular configuration disposed at least partially around the first fluid passage; wherein the first fluid passage is fluidly coupled with the combustion chamber, and the second fluid passage is fluidly coupled with at least one of the combustion chamber or the first fluid passage; wherein the gaseous fuel nozzle assembly includes a steam conduit fluidly coupled with the second fluid passage and extending radially outward from the fuel nozzle body; and wherein the gaseous fuel nozzle assembly includes an outer wall radially spaced from a first end of the fuel nozzle body such that a first outer fluid passage is at least partially defined between the outer wall and the fuel nozzle body; and wherein the steam conduit extends radially outward into the outer wall and axially aft through the outer wall to emit steam into the combustion chamber.

13. A gas turbine engine, comprising: a compressor section, combustion section, and turbine section in serial flow arrangement, with the combustion section comprising: a combustor liner that at least partially defines a combustion chamber; and a gaseous fuel nozzle assembly, comprising: a steam supply to supply steam; a hydrogen fuel supply to supply gaseous hydrogen fuel; and a fuel nozzle body fluidly coupled with the steam supply and the hydrogen fuel supply to provide the steam and the gaseous hydrogen fuel to the combustion chamber to limit a flame speed in the combustion chamber; wherein the fuel nozzle body includes a first fluid passage and a second fluid passage; wherein the first fluid passage is fluidly coupled with the hydrogen fuel supply; wherein the second fluid passage is fluidly coupled with the steam supply and includes an annular configuration disposed at least partially around the first fluid passage; wherein the gaseous fuel nozzle assembly includes an outer wall radially spaced from a first end of the fuel nozzle body such that a first outer fluid passage is at least partially defined between the outer wall and the fuel nozzle body; wherein the first fluid passage includes a first segment that extends through the outer wall to emit the gaseous hydrogen fuel into the combustion chamber; and wherein the second fluid passage includes a second segment that extends through the outer wall and at least partially around the first

segment to emit the steam into the combustion chamber.

14. A gas turbine engine, comprising: a compressor section, combustion section, and turbine section in serial flow arrangement, with the combustion section comprising: a combustor liner that at least partially defines a combustion chamber; and a gaseous fuel nozzle assembly, comprising: a steam supply to supply steam; a hydrogen fuel supply to supply gaseous hydrogen fuel; and a fuel nozzle body fluidly coupled with the steam supply and the hydrogen fuel supply to provide the steam and the gaseous hydrogen fuel to the combustion chamber to limit a flame speed in the combustion chamber; wherein the gaseous fuel nozzle assembly includes a manifold with a mixing chamber fluidly coupled with the fuel nozzle body, the mixing chamber configured to mix the steam with the gaseous hydrogen fuel; and wherein the manifold is separate from the fuel nozzle body; wherein the gaseous fuel nozzle assembly includes: an outer wall radially spaced from a first end of the fuel nozzle body, a first intermediate wall disposed at least partially between the outer wall and the fuel nozzle body, a second intermediate wall disposed at least partially between the first intermediate wall and the outer wall, and wherein the first fluid passage includes a first segment extending through the second intermediate wall to the combustion chamber; wherein the gaseous fuel nozzle assembly includes a first fluid passage fluidly coupled with the hydrogen fuel supply and the steam supply via the manifold; wherein the gaseous fuel nozzle assembly includes a second fluid passage fluidly coupled with the steam supply; and wherein the second fluid passage includes a second segment extending through the first intermediate wall to emit the steam into the combustion chamber.
